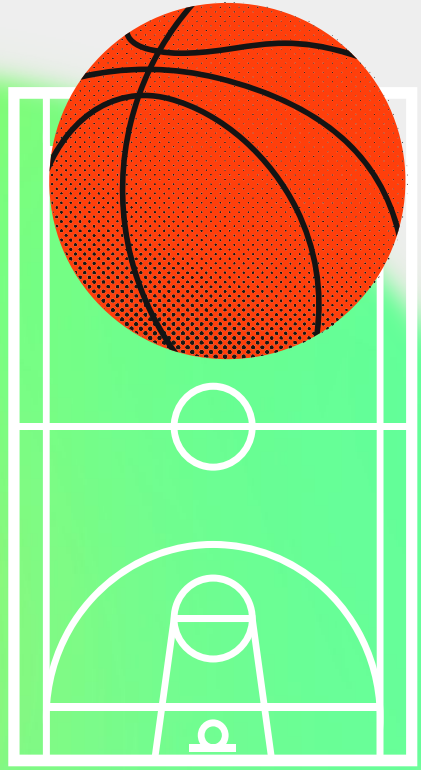
Does Draft Position Predict Career Win Shares in the NBA?



SAL 603 Final Project

Kian Putnam

Introduction

The NBA draft is one of the most important mechanisms for team building in professional basketball. Teams invest significant resources into scouting and evaluating prospects with the goal of maximizing long term player value.

This project investigates whether draft position is actually predictive of long term NBA success. Specifically, I analyze whether draft pick number significantly predicts a player's career Win Shares.

Win Shares is an advanced metric that estimates the number of team wins contributed by a player. It combines offensive and defensive production into one overall value measure. Understanding the strength of this relationship helps evaluate how effective draft positioning truly is as a predictor of player value.

Research Questions

01 Is draft pick significantly correlated with career Win Shares?

02 How much variation in career Win Shares is explained by draft position?

03 Does draft position remain predictive outside of the lottery?

04 Who have been the biggest steals in recent history based on predicted Win Shares?

Data

All data was scraped directly from Basketball Reference draft pages using the pandas read_html() function.

Draft classes from the 2000–2024 NBA seasons were included. All data cleaning and transformations were performed entirely within a Python notebook.

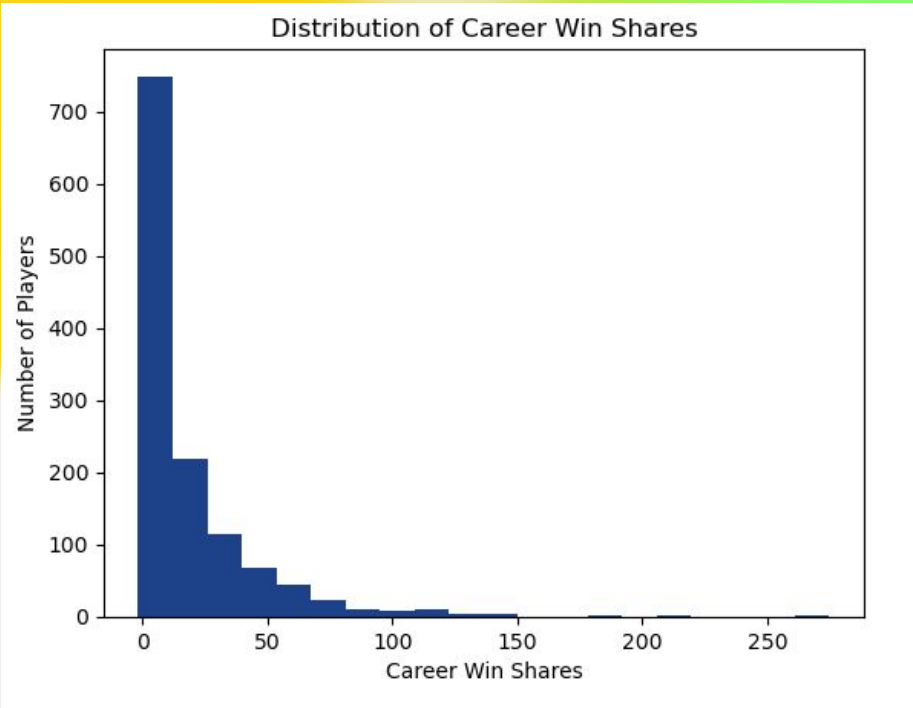
These variables were the main data covered:

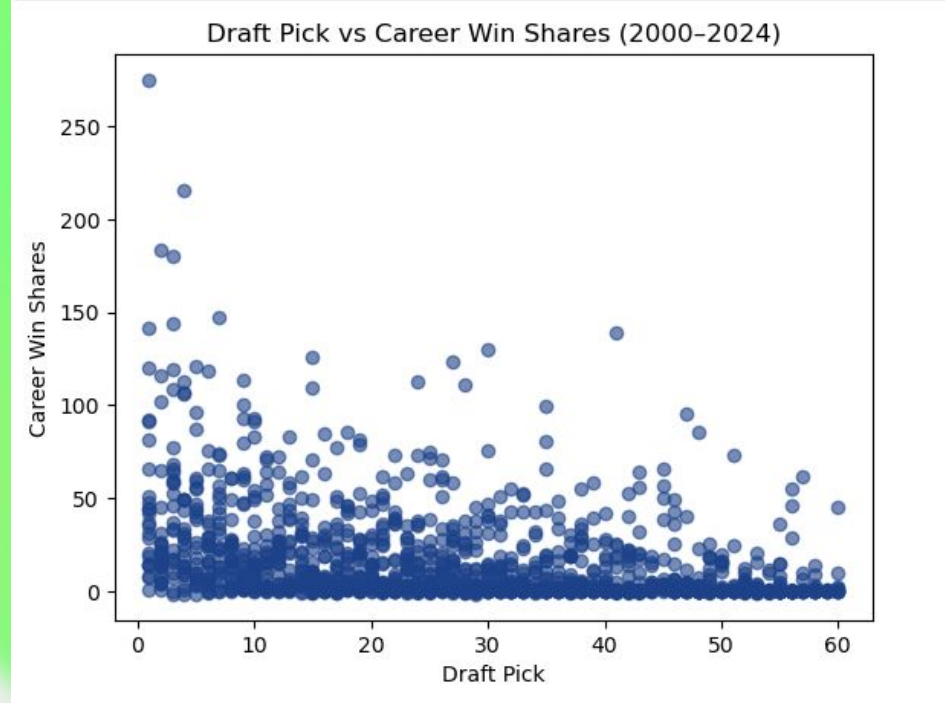
- Pick (Draft Position)
- Player
- Career Win Shares (WS)
- Draft Year

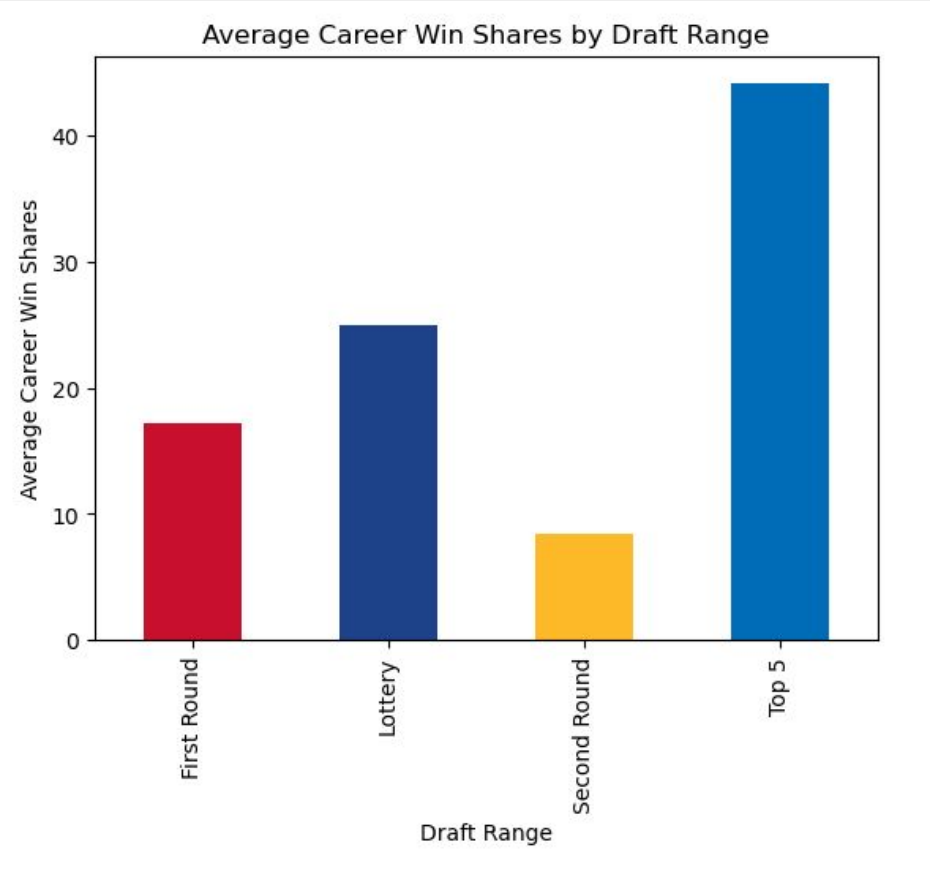
```
draft_df.head()
```

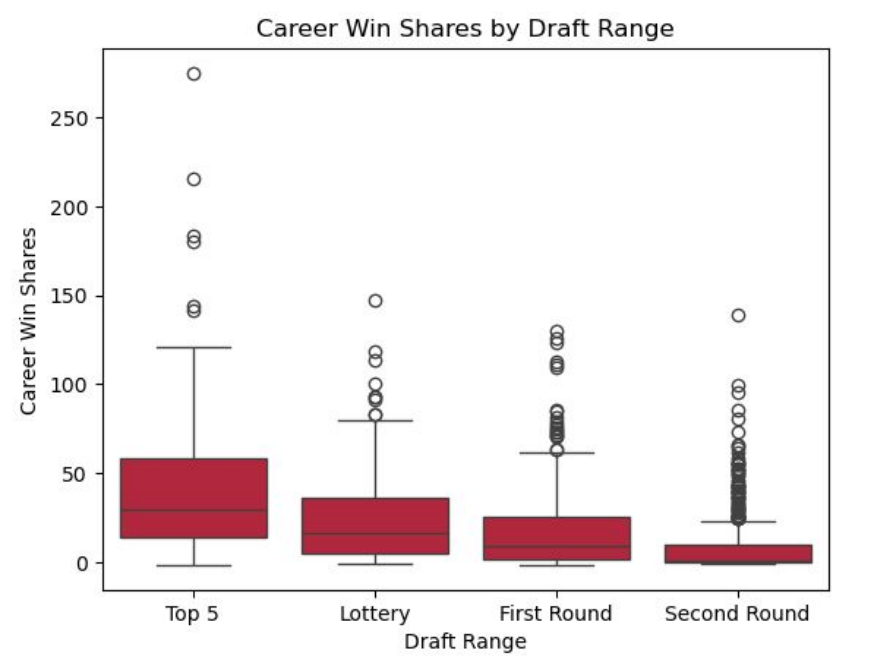
	Rk	Pk	Tm	Player	College	Yrs	G	MP	PTS	TRB	...	FT%	MP.1	PTS.1	TRB.1	AST.1	WS	WS/48	BPM	VORP	DraftYear
0	1	1	NJN	Kenyon Martin	Cincinnati	15	757	23134	9325	5159	...	.629	30.6	12.3	6.8	1.9	48.0	.100	0.1	12.1	2000
1	2	2	VAN	Stromile Swift	LSU	9	547	10804	4582	2535	...	.699	19.8	8.4	4.6	0.5	21.3	.095	-1.6	1.1	2000
2	3	3	LAC	Darius Miles	NaN	7	446	11730	4507	2190	...	.590	26.3	10.1	4.9	1.9	9.5	.039	-1.0	3.0	2000
3	4	4	CHI	Marcus Fizer	Iowa State	6	289	6032	2782	1340	...	.691	20.9	9.6	4.6	1.2	2.7	.022	-3.7	-2.6	2000
4	5	5	ORL	Mike Miller	Florida	17	1032	27812	10973	4376	...	.769	26.9	10.6	4.2	2.6	60.7	.105	0.8	19.8	2000

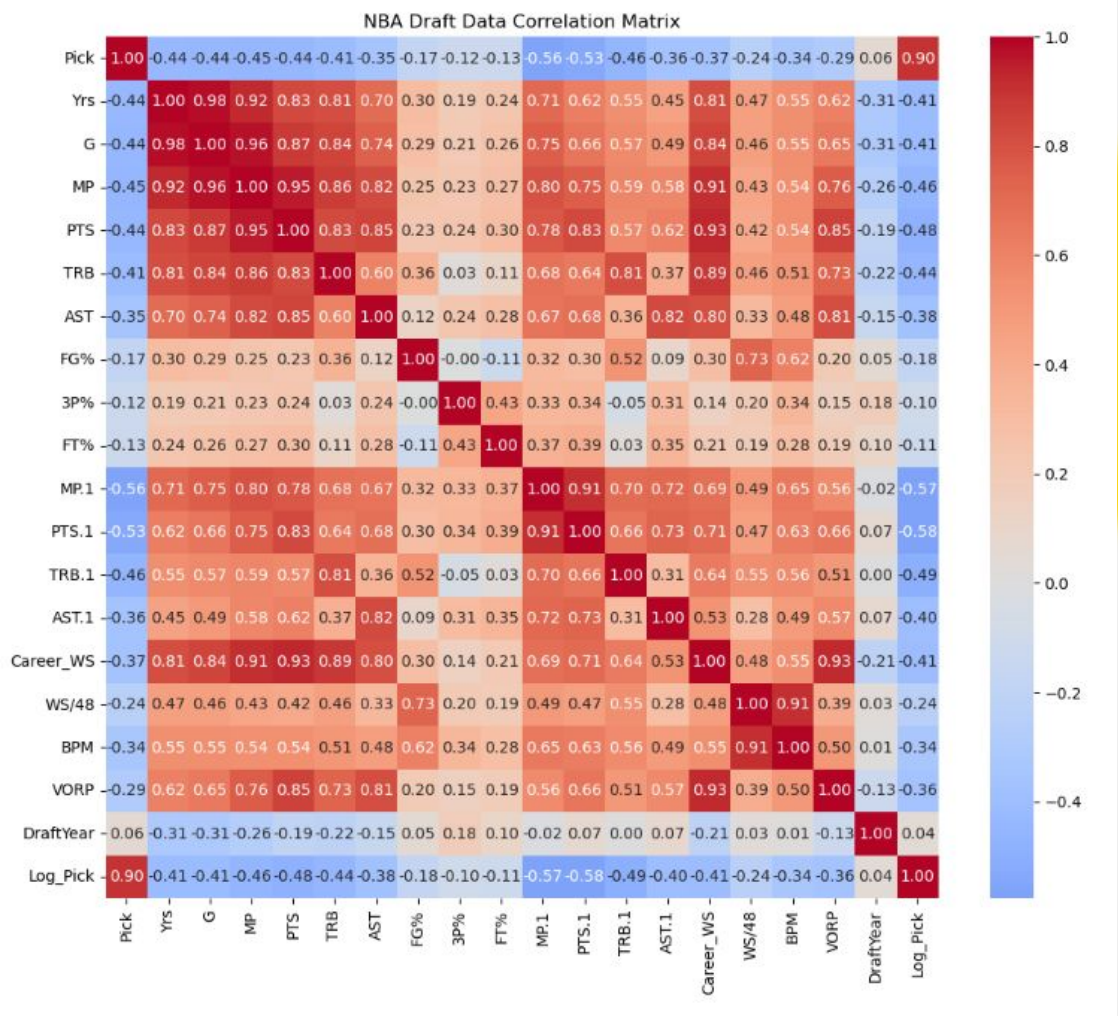
EDA


Distribution of Career Win Shares
Number of Players
Career Win Shares
0
100
200
300
400
500
600
700
0
50
100
150
200
250


Draft Pick vs Career Win Shares (2000–2024)
Career Win Shares
Draft Pick
0
50
100
150
200
250
0
10
20
30
40
50
60


Average Career Win Shares by Draft Range
Average Career Win Shares
0
10
20
30
40
First Round
Lottery
Second Round
Top 5
Draft Range


Career Win Shares by Draft Range
Career Win Shares
0
50
100
150
200
250
Top 5
Lottery
First Round
Second Round
Draft Range



There is a visible negative relationship between draft pick and career Win Shares. Earlier picks tend to produce higher total Win Shares, though there is substantial variability. There is a slight drop off for each step down, from top 5 picks to the lottery, to the rest of the first round.

Correlation

The correlation coefficient between draft pick and career Win Shares is -0.365. The negative sign indicates that lower draft numbers (earlier picks) tend to be associated with higher career Win Shares. Players selected earlier in the draft generally produce more value over the course of their careers. However, the magnitude of the correlation suggests that the relationship is moderate rather than strong. While draft position does appear to matter, it does not fully determine long term player success.

To further evaluate the strength of the relationship, the coefficient of determination (R^2) was calculated. The R^2 value is 0.133, meaning that draft position explains approximately 13.3% of the variation in career Win Shares. This indicates that while draft pick has some predictive power, the majority of variation in player career performance is driven by other factors such as player development, injuries, team fit, etc.

```
Correlation between Draft Pick and Career Win Shares: -0.3651487445719968
```

```
R-squared: 0.13333360566250538
```

Regression

The regression analysis estimates the effect of draft position on career Win Shares.

The coefficient on Pick is -0.5909, indicating that for each one position drop in the draft order, expected career Win Shares decrease by approximately 0.59 wins. Players selected later in the draft tend to contribute fewer wins over the course of their careers.

The regression results also show that draft position is statistically significant. The p-value associated with the Pick coefficient is less than 0.001, meaning the relationship between draft position and career Win Shares is highly unlikely to be due to random chance.

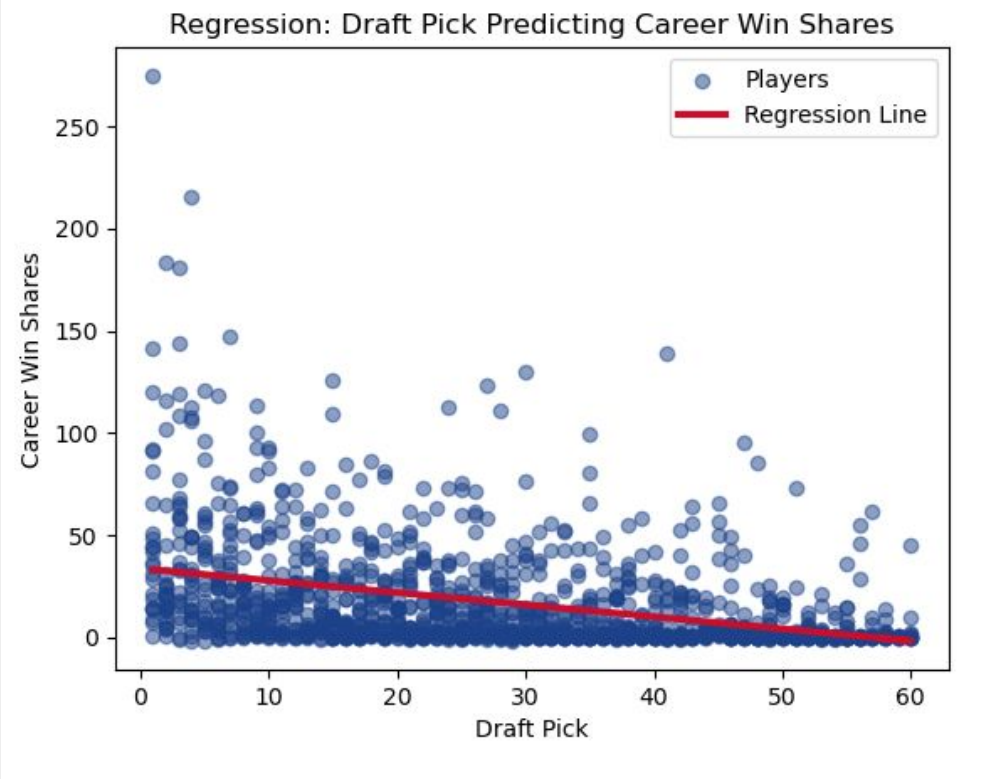
Again however, the R-squared value is 0.133, which indicates that draft position explains only about 13.3% of the variation in career Win Shares. This suggests that while draft position is a meaningful predictor of long term performance, the majority of variation in player career value is driven by other factors not captured in this model.

Overall, the regression results support the idea that earlier draft picks tend to produce more career value, but they also highlight that draft position alone is far from a complete predictor of NBA success.

Regression cont.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               Career_WS   R-squared:                       0.133
Model:                            OLS   Adj. R-squared:                  0.133
Method:                 Least Squares   F-statistic:                     192.8
Date:                Thu, 05 Mar 2026   Prob (F-statistic):           7.12e-41
Time:                        20:22:52   Log-Likelihood:                -5805.0
No. Observations:                1255   AIC:                         1.161e+04
Df Residuals:                    1253   BIC:                         1.162e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         33.6231      1.367     24.594      0.000      30.941      36.305
Pick          -0.5909      0.043    -13.884      0.000      -0.674      -0.507
==============================================================================
Omnibus:                      838.742   Durbin-Watson:                   1.796
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            12925.299
Skew:                           2.899   Prob(JB):                         0.00
Kurtosis:                      17.614   Cond. No.                         63.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Lottery Vs Non Lottery

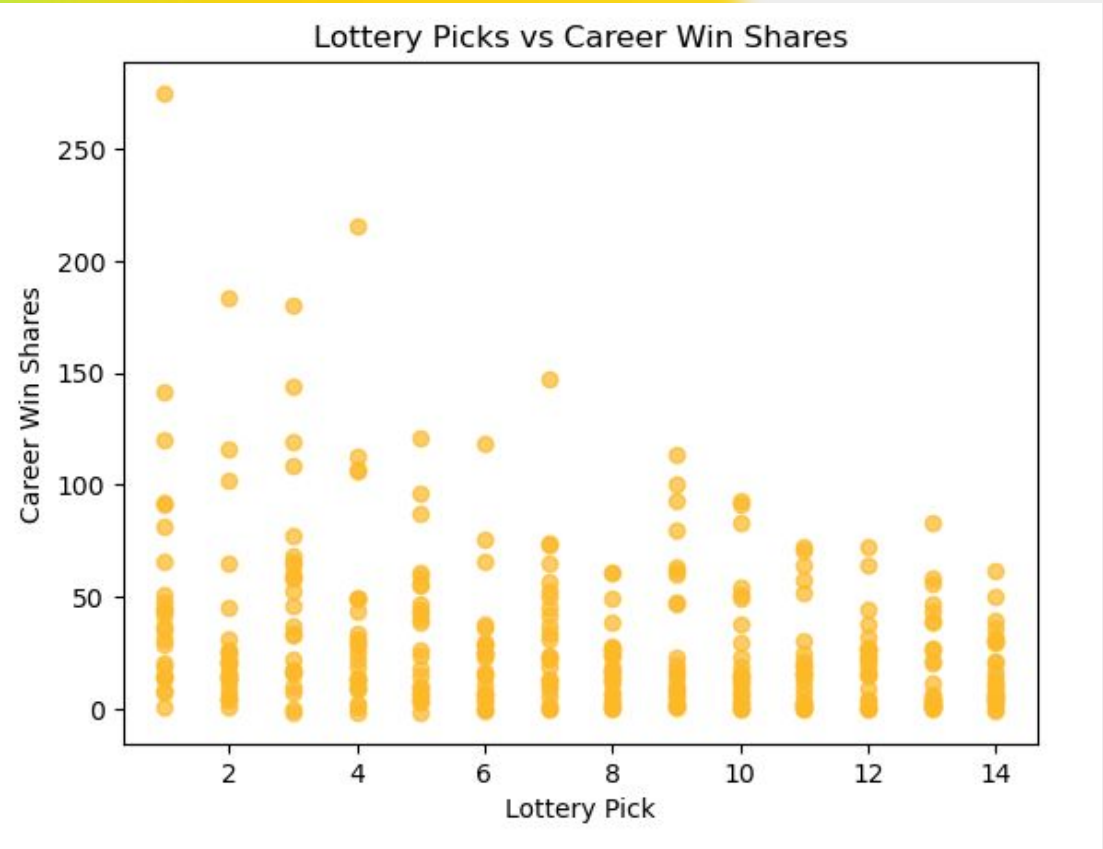
I next divided the dataset into lottery picks (1–14) and non-lottery picks (15 and later) to examine whether draft order is more predictive among the highest valued prospects or whether the relationship remains consistent across the entire draft.

The correlation between draft pick and career Win Shares among lottery selections is -0.265, indicating a modest negative relationship. This suggests that even among the top prospects, players selected earlier tend to generate slightly more career value than those selected later within the lottery. Among non-lottery selections, the correlation is -0.233. This relationship is slightly weaker but still negative, indicating that draft position continues to have some predictive value even outside the lottery.

Overall, both correlations are weaker than the full-sample correlation, suggesting that while earlier picks generally perform better, the relationship becomes less pronounced when examining smaller portions of the draft. This highlights the increasing uncertainty in projecting long-term player value once teams move beyond the very top of the draft.

```
Lottery correlation: -0.2648008096525358
Non-lottery correlation: -0.23324096898094537
```

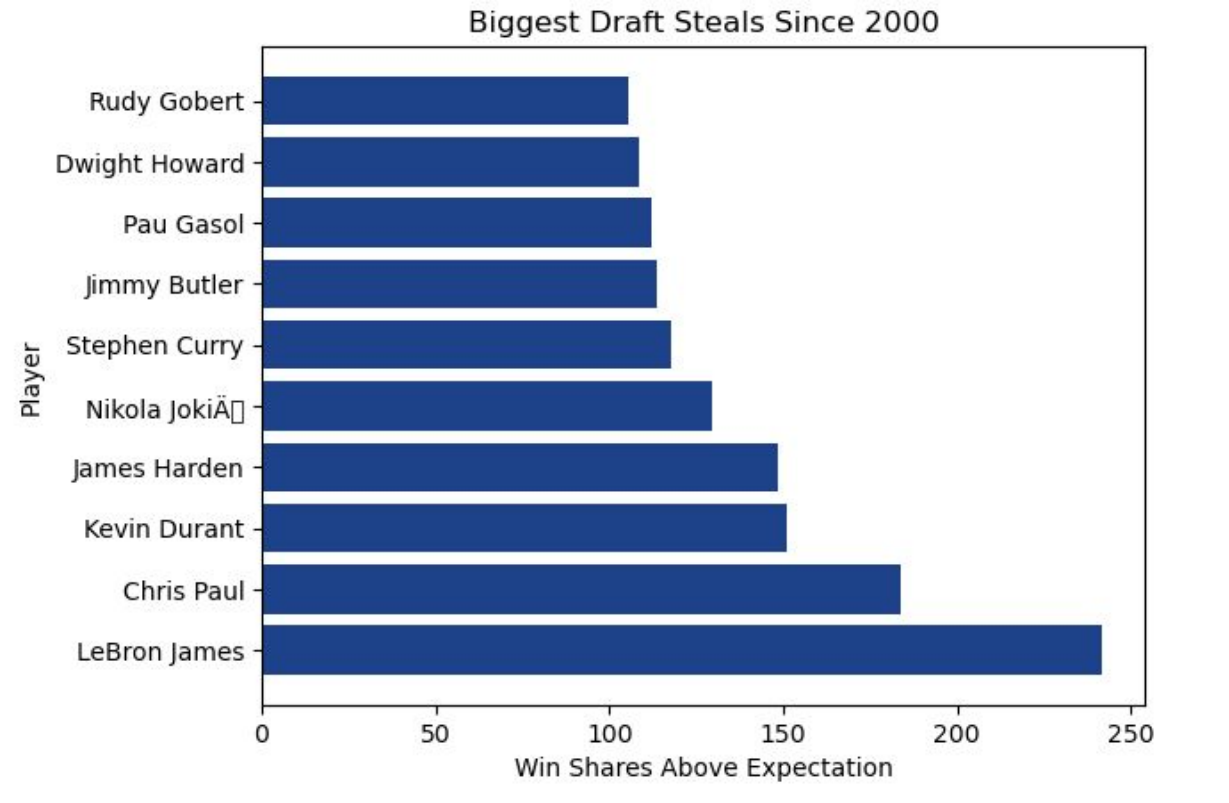
Lottery Vs Non Lottery Cont.




```
draft_df.groupby("Pick")["Career_WS"].std()
```

```
Pick
1.0     59.304817
2.0     43.012511
3.0     46.164299
4.0     49.358047
5.0     32.388752
6.0     27.633439
7.0     32.980306
8.0     17.541115
9.0     35.490549
10.0    28.952322
11.0    23.810113
12.0    18.637036
13.0    23.369685
14.0    16.946527
15.0    33.964188
16.0    21.158135
17.0    20.165310
18.0    21.474484
19.0    22.831252
20.0    13.203982
21.0    19.826830
22.0    21.469350
23.0    14.714204
24.0    26.736686
25.0    22.823970
26.0    21.860794
27.0    26.485153
28.0    24.150317
29.0    13.972060
30.0    31.400499
31.0    14.597352
32.0    13.986727
33.0    16.583838
34.0    12.508163
35.0    28.912993
36.0    14.738525
37.0     9.985974
38.0    15.474677
39.0    16.663731
40.0    11.460470
41.0    29.816830
42.0    14.746817
43.0    19.164701
44.0     5.989654
45.0    21.827548
46.0    15.980364
47.0    23.989986
48.0    19.307715
49.0     8.535019
50.0     7.741281
51.0    20.022657
52.0     3.781106
53.0     5.863234
54.0     4.066215
55.0     9.203333
56.0    18.596083
57.0    19.059815
58.0     4.127181
59.0     0.222539
60.0    14.145608
Name: Career_WS, dtype: float64
```

Draft Steals



Players with the highest positive residuals significantly outperformed what their draft position predicted. These players represent the biggest draft steals in the dataset.

Next players: Giannis Antetokounmpo, Tony Parker, Kyle Lowry, Dwyane Wade, Paul Millsap, Damian Lillard, Al Horford, Anthony Davis, Deandre Jordan, Demar Derozan

Conclusion

This analysis examined whether NBA draft position is predictive of long term player success, measured by career Win Shares. The results show a statistically significant negative relationship between draft position and career Win Shares. Earlier draft picks tend to produce higher overall player value, while players selected later in the draft generally accumulate fewer career Win Shares.

However, the strength of this relationship is moderate rather than strong. The correlation between draft position and career Win Shares was approximately -0.37, indicating a meaningful but not overwhelming association. The regression analysis produced an R-squared value of about 0.13, meaning that draft position explains only about 13% of the variation in career Win Shares. This suggests that while draft order does provide useful information about expected player value, the majority of career outcomes are influenced by other factors such as player development, injuries, team context, and opportunity.

Conclusion Cont.

When examining draft position within subsets of the draft, the relationship becomes slightly weaker. The correlation among lottery picks was approximately -0.26, while the non-lottery picks had a correlation of about -0.23. This indicates that although earlier selections within these groups still tend to perform better, outcomes become more unpredictable as teams move further down the draft board.

Additionally, the analysis of regression residuals revealed several players who significantly exceeded or fell short of expectations based on their draft position. These cases highlight the presence of both draft steals and underperforming high picks, reinforcing the idea that evaluating basketball talent involves substantial uncertainty.

Overall, the results suggest that draft position does matter and is statistically related to long-term player performance. However, it is far from a perfect predictor of success. While teams may improve their odds of selecting productive players with earlier picks, the NBA draft remains an inherently uncertain process where unexpected outcomes are common.

Data Source and Acknowledgements

Data Source

- Draft data scraped from **Basketball Reference**
- Draft classes from **2000–2024**

Presentation Template

- PowerPoint template provided by **Slidesgo**

Project Github Repo

- Kianjputnam/SAL603-FProject